Instructions

Dear Valued Customer,

Thank you for choosing Bio Basic Inc. for your gene synthesis services. Here are some electronic files regarding your target gene including our QC report. Should you have any questions or concerns, please do not hesitate to contact us. We would be more than happy to assist you to the best of our ability with your research.

Yours faithfully,

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Instructions for files in the pack:

1. ".pdf". Portable Document Format (PDF) is an open standard for documents exchange. Any pdf viewer software can be used to view this file, or download Adobe® Reader® at: http://www.adobe.com/products/reader.html

2. ".ab1". Sequence trace files are the output files from Applied Biosystem's Sequencing Analysis software. Any chromatograms viewer or download "Chromas Lite" at: http://www.technelysium.com.au/chromas_lite.html

3. ".fasta". Fasta format is a text-based format, a sequence inf fasta format begins with a single-line description, followed by lines of sequence data. The description line is distinguished from the sequence data by a greater-than (">") symbol in the first column. The file can be viewed and edited with ease using a text editor.

4. ".seq". Sequence file created by a sequence analysis software called Lasergene®, which can be downloaded at: http://http://www.dnastar.com/, this file can also be viewed simply by any text editor.

5. ".sqd". Sequence alignment by Lasergene®, If you do not have this software to view alignment, a pdf file show alignment results is also provided.

Instructions for plasmid reconstitution:

1. Before opening the tube containing the DNA, please briefly centrifuge the tube. Lyophilized DNA could attach to the wall of the tube. Opening without centrifugation may cause DNA loss.

2. The lyophilized plasmid is stable at -20°C for at least 1 year. Plasmid dissolved in TE is stable for at least 6 months at -20°C or 4°C. Plasmid dissolved in water is stable for at least 6 months at -20°C in the absence of nucleases. Be sure the water used is at neutral pH to avoid depurination.

Recommended protocols for re-plating:

1. Re-suspend lyophilized DNA (4 µg) in 40 µl of 10 mM Tris buffer (pH 8.5). Final concentration is ~100 ng/µl. This is the original stock. Please note, 4 µg is an approximate amount. To accurately determine the quantity of DNA present, please measure by optical density at OD260.

2. In a new centrifuge tube, make a 1:10 dilution of original stock in 10 mM Tris buffer (pH 8.5). This is 1:10 diluted stock.

3. To plate DNA, transform 2 µl of original stock DNA and 2 µl 1:10 diluted stock DNA into appropriate *E. coli* competent cells. Incubate mixture on LB agar plates (with desired antibiotic selection) at 37°C overnight.

4. Select a well separated, **SINGLE** colony and inoculate in LB medium with desired antibiotic selection for larges scale culture. **NOTE: It is important to select a SINGLE colony.**

5. Purify DNA from large culture. Verify sequences by sequencing and continue with project of interest.

6. If you encounter any problems, please do not hesitate to contact us. We will be more than happy to assist you.

Instructions for bacteria growth:

If not stated, *E. coli* cells harboring plasmids are cultured in LB medium with appropriate antibiotic at 37°C during our manipulation.